fil_001

To submit a proposal, please create a PR against this template in this repo. Please title your file `open-proposal-title.md`, replacing `title` with the name of your project.

Open Grant Proposal: `Lotus Farcaster`

Name of Project: `lotus farcaster V3`

Proposal Category: Choose one of `app-dev`

Proposer: `s0nik42 - Twin Quasar`

Do you agree to open source all work you do on behalf of this RFP and dual-license under MIT and APACHE2 licenses?: “Yes”

Project Description

lotus-farcaster V3 brings additionnal functionnalities to lotus farcaster (see tables below).

Lotus farcaster is used by 400+ miners, was downloaded more than 200 times and get 30+ new users during the last 2 weeks. We want to keep farcaster has one of the main analytics companion of miners by continuously adding features supporting their new challenges. This new release follow Filecoin roadmap and bring storage markets features, bring support to the new lotus split mode architecture and made it reusable for other project by providing all the functions are a python module. This tool is build for the community and we really want to keep it opensource

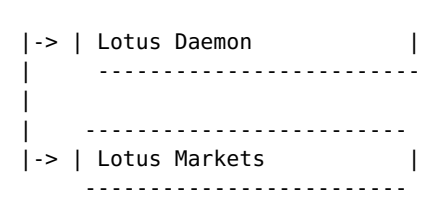
Lotus farcaster is a Visualization and Analytics tool for Lotus Filecoin node It leverages Prometheus, Grafana and Python.

We want it to be accessible and designed for any types of miners. It’s designed to manage all type of architecture : * single miner, * single miner with workers * monolitic and split store architecture (markets node) * multiple miners farm.

We want to move from console based monitoring to a proactive professional monitoring grade. Compared to the existing console tools, users will benefits : * consolidated information * data history * push notification * Visual monitoring * Scalable solution * User management * Security * Mobile device accessibility

How it works

Architecture



lotus farcaster comes with 2 Components :

- A Grafana dashboard
- lotus-exporter-farcaster a standalone python script executed every minute by the crontab. It generates metrics that are exposed by node-exporter to prometheus.

Security

The design and choice of these components, was made to keep a high level of security : - metrics are pulled from the miner to avoid compromising network or management servers - lotus-exporter-farcaster doesn’t bind any ports to avoid exposing the miner to external threat - lotus-exporter-farcaster run under unprivileged user (ideally lotus user) to avoid system being compromised - The global solution doesn’t rely on any cloud or remote component. To avoid data leak.

lotus-exporter-farcaster is running locally on the miner without any specific privilege nor access to other servers.

It will follow **prometheus guidelines** :

- ressources inexpensive
- standard prometheus metrics naming
- compatible with 3rd parties dashboards and exporters

Benefits

- Support any type of lotus architecture
- Easy to deploy
- Small footprint
- Collect lotus node and miner data
- Only rely on API
- Data are pulled from the Prometheus server (increase security)
- Deployed on the miner node only
- Run under Unprivileged user

Dashboard Features

Feature	Status	Comments
Sectors	V1	
Mpool	V1	
Storage Info	V1	
Workers Info	V1	
Sealing	V1	Job + Scheduler
Power	V1	
Wallets	V1	
Chain	V1	Head + Sync Status
Deadlines	V1	
Deals	V2	View of current sealing deals (State/Price/Size/collaterals/From/Is Verified/etc…
Gas Fee	V2	Realtime gasfee and History (requested by Community)
Sectors State	V2	Identify faulty sectors in deadlines (requested by Community). Helping miners to quickly indentify which

		sectors to restore
Fil+ Datacap	V2	Fil+ Realtime and Historical view of Datacap allocated to the miner addresses
Max Quality Adjusted Power	V2	A new indicator, showing the maximum power that can be expected from a miner (it’s a forecast of the power after the next deadline if the full capacity is proved)
Sectors w/ Deals	V2	Breakdown of sectors in 3 categories : CC / inc. deals / inc. Verified. Current and historical view, helping the miner to understand how he gets is power over time
Address lookup	V2	Allow miners to provide a custom lookup file (Adress=Name/Label) to improve dashboard readability.. The lookup apply to any panels containing adresses (Wallet / Deals / Fil+ / etc…).
Farcaster Status	V2	Farcaster reports any problems when scraping data. Ensuring dashboard is accurate.
Deployment toolset	V3	Provide a deployement script supporting and autodetecting any type of lotus architecture + Docker
Full Python Library	V3	Provides all functionnalities as a Python library.
Storage market Proposal	V3	Provides on-going downloads, pending publishing deals, etc…
Sectors lifecycle	V3	Dashboards helping miners to identify expired sectors correlated with errors to help fast troubleshooting.

Roadmap

The planning could be shorted if Protocol Labs for any reasons decide to force miners to switch to splitstore.

01 sep 2021 : V3 scoping 07 sep 2021 : V3 Development of Design / Specifications / Mockup 30 sep 2021 : First release open to beta tester 15 oct 2021 : Public release of the new farcaster (without the python lib) 15 nov 2021 : Public release of the python lib

Milestone	Hours	Cost
Collect needs from the community, design, planning	8	€ 576
Studing the new Lotus API and reverse iengineering	8	€ 576
Exporter development (lotus V3.0)	40	€ 2 880
Dashboard development (lotus V3.0)	40	€ 2 880
Collect feedbacks from beta user / bug fix and UI	10	€ 720
Automatic Deployment process + Docker	20	€ 1 440
Python library	50	€ 3 600
Launch support (community support / comm / bug fix)	20	€ 1 440
New Video Tutorial + Documentations (Docstring + Dashboard request)	20	€ 1 440

First year support (up to 8 hours)	8	€ 576
Total Budget Requested		**€ 16 128**
Option : Additionnal support per major update (start after included support consummed)	10	€ 720

No VAT charged

Maintenance and Upgrade Plans

Tools will be updated according to lotus release. Will be glad to work closer with PL team to specify API needs.

Team

Team Members

Julien NOEL - julien.noel@twinquasar.io

Team Website

`https://twinquasar.io`

Relevant Experience

Julien is a : - Filecoin miner, actively helping the filecoin community on slack. - Filecoin Notary - Part of MinerX program IPFS and Web3 enthousiast, starting being active around filecoin right before Space Race, Julien is a computer science engineer, spending 15 years working for Top100 companies providing strategic advice to C-level around new technologies.